Isabela Garza

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EDUCATION

Boston University, Boston, MA - *Master of Computer Science and Artificial Intelligence* Expected Dec 2027

The University of Texas at Austin, Austin, TX - *Bachelor of Science, Computer Science* May 2026
Minor: Robotics
Relevant Coursework: Operating Systems, Data Structures, Linear Algebra, Statistics, Machine Learning, Algorithms, Artificial Intelligence, Computer Vision

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, Swift, SwiftUI
Data Science & ML: pandas, NumPy, scikit-learn, SciPy,
Machine Learning: Classification, feature engineering, reinforcement learning, model evaluation
Tools: Git, Linux, Firebase, OpenCV, NVIDIA Isaac Lab, OpenAI, GitHub Copilot, ChatGPT, Claude

RESEARCH EXPERIENCE

Undergraduate Research Assistant - Human-Centered Robotics Lab
Jun – Aug 2025 | The University of Texas at Austin
- Conducted research on reinforcement learning–based control for a multipedal robot operating in zero-gravity environments by developing and training policies using Isaac Lab (NVIDIA Isaac Sim) for physics-based simulation
- Simulated smooth reorientation and manipulation behaviors for robotic vessel repair tasks
- Evaluated policy performance across stability, efficiency, and task success metrics

DATA SCIENCE & ML PROJECTS

SoundVault (Swift) Fall 2025
- Designed and developed a full-stack iOS application that allows users to search, rate, and review music albums
- Integrated the Spotify API and translated album data into intuitive user-facing features
- Implemented user authentication, user profiles, ratings, reviews, and cloud storage using Firebase
- Iterated on application interface behavior based on usability testing

Student Struggle Prediction Model (Python, pandas, scikit-learn) Summer 2025
- Conducted exploratory analysis and engineered behavioral features to predict student learning difficulties
- Trained and evaluated logistic-regression models, improving ROC-AUC from 0.829 to 0.851
- Analyzed model performance and feature relationships to identify signals associated with student struggle

Austin Pets Alive! Outcome Predictor ML Model (Python, pandas, scikit-learn) Spring 2025
- Cleaned and analyzed real-world animal shelter data, addressing missing values and preparing categorical and numerical features for modeling
- Engineered meaningful features using domain insights to enhance predictive performance
- Trained and compared multiple ML models (Random Forest, Decision Tree, Neural Network, XGBoost), improving classification accuracy

Cloud Scheduling Simulator (C++) Fall 2024
- Developed scheduling algorithms to optimize energy consumption and hardware longevity under varying workloads
- Modeled dynamic voltage and frequency scaling, load balancing, and power-gating decisions in a cloud-computing simulation
- Ran experiments and analyzed trade-offs among energy efficiency, system utilization, performance, and reliability

TEACHING EXPERIENCE

Undergraduate Course Assistant
Aug 2024 - Jun 2026 | The University of Texas at Austin
- Communicated complex programming and algorithmic concepts to classes of more than 200 students
- Conducted office hours to tend to student needs, clarify ambiguous or complex problems, and guide students towards workable solutions
- Collaborated with professors and other course assistants to identify recurring challenges and improve instructional support